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ACT SUBJECT TEST — MATH 2 • UNIT 6.01

Limits, Asymptotes & Instantaneous Rates

PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Limits, Asymptotes & Instantaneous Rates Practice Exercises

1 MCQ

The graph of the function $f(x) = (x-1)^2 - 4$ is shown in the standard (x, y) coordinate plane below. What is the range of f?

A $[-4, \infty)$

B $[1, \infty)$

C $(-\infty, -4]$

D $(-\infty, \infty)$

2 MCQ

For the complex number $i = \sqrt{-1}$, which of the following is equivalent to the expression $(5 - 3i) - (2 - 7i)$?

A $3 + 4i$

B $3 - 10i$

C $7 + 4i$

D $3 - 4i$

3 MCQ

What is the value of $\lim_{x \to 3} \frac{x^2 - 2x - 3}{x - 3}$?

A 4

B 0

C 6

D The limit does not exist

4 MCQ

Given the vectors $\mathbf{u} = \langle 3, -4 \rangle$ and $\mathbf{v} = \langle -2, 5 \rangle$, what is the vector $2\mathbf{u} + \mathbf{v}$?

A $\langle 4, -3 \rangle$

B $\langle 1, 1 \rangle$

C $\langle 4, 1 \rangle$

D $\langle 8, -13 \rangle$

5 MCQ

The first four terms of an arithmetic sequence are 5, 12, 19, and 26. What is the 20th term of this sequence?

A 131

B 138

C 140

D 145

6 MCQ

A point has polar coordinates $(r, \theta) = (6, \frac{2\pi}{3})$. What are the Cartesian coordinates (x, y) of this point?

A $(-3, 3\sqrt{3})$

B $(-3\sqrt{3}, 3)$

C $(3, -3\sqrt{3})$

D $(3\sqrt{3}, -3)$

7 MCQ

The polynomial function $P(x) = 2x^3 + ax^2 - 5x + b$ has $(x - 1)$ as a factor. When $P(x)$ is divided by $(x + 2)$, the remainder is -18. What is the remainder when $P(x)$ is divided by $(x - 3)$?

A 2

B 8

C 18

D -30

8 MCQ

For what real value of k is the complex expression

$$\frac{k + 8i}{3 - i} + \frac{2 - ki}{3 + i}$$

a real number?

A 11

B 1

C 6

D -11

9 MCQ

For real constants a and b, suppose that

$$\lim_{x \to 0} \frac{\sqrt{ax + b} - 3}{x} = 2.$$

What is the value of $a + b$?

A 12

B 15

C 21

D 27

10 MCQ

Let $P(x)$ be a polynomial of minimum degree with real coefficients such that

$$\lim_{x\to 1}\frac{P(x)}{x^2-3x+2}=5 \quad \text{and} \quad \lim_{x\to 2}\frac{P(x)}{x^2-3x+2}=-3.$$

What is the value of $P(3)$?

A -11

B -22

C 22

D 42

11 MCQ

What is the value of the limit $\lim_{x\to 2}\frac{x^2+x-6}{x-2}$?

A 5

B 1

C -1

D Does not exist

12 MCQ

If the limit $\lim_{x\to 3}\frac{x^2+kx-15}{x^2-4x+3}=L$ exists and is a finite real number, where k and L are constants, what is the value of $k+L$?

A 4

B 6

C 8

D 2

13 MCQ

For each real number $t\in(-2,2)$, the cubic equation $x^3-3x+t=0$ has three distinct real roots, denoted in increasing order by $x_1(t)<x_2(t)<x_3(t)$.
What is the value of the limit

$$\lim_{t\to 2^-}\frac{x_3(t)-x_2(t)}{\sqrt{2-t}}\ ?$$

A $\sqrt{3}$

B $\frac{2\sqrt{3}}{3}$

C $\frac{\sqrt{3}}{2}$

D $\frac{4}{3}$

14 MCQ

What is the value of $\lim_{x\to 2}\frac{x^2+x-6}{x-2}$?

A 5

B 1

C 0

D Does not exist

15 MCQ

For real numbers a and b, suppose that

$$\lim_{x \to 1} \frac{x^2 + ax + b}{x^2 - 1} = 3.$$

What is the value of $a - b$?

A -1

B 9

C 3

D -9

16 MCQ

In the xy-plane, let $P(t, t^2)$ with $t > 0$ be a point on the parabola $y = x^2$. The normal line to the parabola at P intersects the parabola again at a second point Q. Let T be the intersection point of the tangent lines to the parabola at P and Q. As $t \to 0^+$, what is the limiting value of the y-coordinate of T?

A $-\frac{1}{4}$

B -1

C 0

D $-\frac{1}{2}$

17 MCQ

What is the value of $\lim_{x \to 3} \frac{2x^2 - 5x - 3}{x - 3}$?

A 7

B 5

C 1

D 0

18 MCQ

Find $\lim_{x \to 4} (-3x^2 - 6)$.

A -54

B -53

C -56

D 54

19 MCQ

Find $\lim_{x \to -3} (-3x^2 + 4x + 3)$.

A -35

B -38

C -36

D 36

20 MCQ

Find $\lim_{x \to -3} (x^2 - 4x)$.

A 22

B 19

C 21

D -21

21 MCQ

Find $\lim_{x \to 3} (-4x^2 + 4x + 4)$.

A -20

B -19

C -22

D 20

22 MCQ

Find $\lim_{x \to -2} (-3x^2 + 2x - 6)$.

A -21

B -22

C -24

D 22

23 MCQ

Find $\lim_{x \to 4} (-2x^2 - 3x - 4)$.

A -50

B 48

C -47

D -48

24 MCQ

Find $\lim_{x \to 0} (-2x^2 - 2x)$.

A 0

B 1

C -2

D 5

25 MCQ

Find $\lim_{x \to 1} (4x^2 - 6x - 2)$.

A -4

B -3

C -6

D 4